

Moyang Wang

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Madison, WI

Education

University of Wisconsin–Madison

B.S. in Computer Science, in progress

Madison, WI

GPA: 3.74/4.00

Relevant coursework: CS 532: Matrix Methods in Machine Learning; CS 540: Introduction to Artificial Intelligence; STAT 453: Introduction to Deep Learning and Generative Models.

Independent study: CS 639: Foundation Models (completed self-study); CS 639: Introduction to Reinforcement Learning (self-study).

Research Interests

Language model training and post-training; representation editing and activation steering; parameter-efficient adaptation; efficient inference and state management; multimodal foundation models.

Research Experience

State-Aware Adaptation and Control of Language Models

Jun. 2026 – Present

Independent research

- Developed experimental infrastructure to separate the effects of current model parameters from cached computation, using a four-condition design crossing base/adapted readers with base/adapter-written KV caches.
- Implemented activation-steering schedules and fixed-token counterfactual evaluations; added switch-off, random-direction, orthogonal-direction, and numerical-precision controls to distinguish state effects from generated-token feedback.
- Analyzed steering withdrawal on Qwen and Llama models, including next-token alignment and refusal behavior; documented negative results and limits on attributing residual changes to the intended steering direction.
- Investigating lightweight KV-cache correction for model adaptation, with an emphasis on quality–compute trade-offs and reproducible evaluation.

Knowledge Graph Construction for Computer Organization

2023 – 2024

Undergraduate research project, Beijing University of Technology

- Co-developed a research proposal for a course-specific knowledge graph, organizing textbook concepts and their hierarchical relations into a domain ontology.
- Designed a semi-automated workflow for educational-text preprocessing, entity recognition with BiLSTM-CRF, and graph storage in Neo4j; reviewed knowledge-extraction approaches, including BERT-based entity prediction.

Selected Project

Decoder-Only Language Model: Dense and Sparse-MoE Variants

2026

Self-directed implementation based on Stanford CS336 Assignment 1

- Built a PyTorch Transformer implementation with causal multi-head attention, RoPE, RMSNorm, SwiGLU, and embedding/output weight sharing; extended the project with configurable dense and top- k sparse-MoE feed-forward layers.
- Completed byte-level BPE, stable cross-entropy, AdamW, learning-rate scheduling, gradient clipping, checkpointing, and a reproducible training CLI; the portfolio release includes AI-assisted completion and extensions.
- Validated the release with **52 passing tests** covering reference outputs, tokenization, optimizer behavior, causality, MoE dispatch/gradients, and shared-weight serialization; two POSIX memory-limit tests are skipped on Windows.

Technical Skills

Programming: Python, C, Java. **ML:** PyTorch, NumPy, Hugging Face Transformers, LoRA, model evaluation.

Engineering: Git, Linux, pytest; Transformer implementation, activation hooks, KV-cache experiments.

Outside research: Calisthenics and gymnastics rings.